

Part A: Literature Review and Public Dataset

CAP182 Capstone Project: STADIOEquities

Publication 1

Requirement	Information
Title	A data-driven approach to predict the success of bank telemarketing
Harvard reference	Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', Decision Support Systems, 62, pp. 22-31. doi: 10.1016/j.dss.2014.03.001.
Link	https://doi.org/10.1016/j.dss.2014.03.001
Modelling	The study compared logistic regression, decision trees, neural networks and support vector machines (SVM). Model performance was evaluated using the area under the ROC curve (AUC) and cumulative LIFT curves. A rolling time-window evaluation approach was also used.
Preprocessing techniques	The study incorporated client demographic information, campaign-contact information and macroeconomic variables. A rolling time-window approach was used to evaluate the models using data from different periods.
Datasets used	Portuguese bank telemarketing campaign data. The dataset was subsequently made publicly available through the UCI Machine Learning Repository as the Bank Marketing dataset.
Key takeaways	1. Customer and campaign-related information can be useful for predicting bank marketing outcomes. 2. Time-aware evaluation is important when working with data collected over different periods. 3. Several different modelling techniques should be compared rather than relying on a single model.

Publication 2

Requirement	Information
Title	Investigating customer churn in banking: a machine learning approach and visualization app for data science and management
Harvard reference	Singh, P.P., Anik, F.I., Senapati, R., Sinha, A., Sakib, N. and Hossain, E. (2024) 'Investigating customer churn in banking: a machine learning approach and visualization app for data science and management', Data Science and Management, 7(1), pp. 7-16. doi: 10.1016/j.dsm.2023.09.002.
Link	https://doi.org/10.1016/j.dsm.2023.09.002
Modelling	The study used grid-search cross-validation to investigate logistic regression, decision tree, random forest and gradient boosting models. A visualisation application was also developed to support the interpretation and use of the results.
Preprocessing techniques	The study used SMOTE to address class imbalance. Customer information from different account-related variables was combined, including credit score, tenure, balance, number of products and estimated salary.
Datasets used	A Kaggle bank customer churn dataset containing 10,000 customer records. The dataset included variables such as credit score, age, tenure, balance, number of products and salary.
Key takeaways	1. Class imbalance should be considered when developing models for binary banking outcomes. 2. Customer account and demographic characteristics can provide useful predictive information. 3. Model results can be presented through visualisation tools to make the findings easier to interpret and communicate.

Publication 3

Requirement	Information
Title	Machine Learning-Based Analysis of Technology Acceptance in FinTech: A Behavioral Study Using Digital Wallet Data
Harvard reference	Abbas, S.K., Hussain, M. and Rimal, Y.N. (2025) 'Machine learning-based analysis of technology acceptance in FinTech: a behavioral study using digital wallet data', SN Computer Science, 6, article 674. doi: 10.1007/s42979-025-04214-8.
Link	https://doi.org/10.1007/s42979-025-04214-8
Modelling	The study used logistic regression, support vector machines (SVM), random forest and XGBoost to classify high and low technology acceptance. The models were re-evaluated after applying SMOTE. SHAP and principal component analysis (PCA) were used to support model interpretation.
Preprocessing techniques	Categorical variables were label encoded, while numerical variables were standardised using z-score standardisation. Identifier columns were removed, and an 80/20 stratified train-test split was used. The dataset contained no missing values.
Datasets used	The Kaggle Digital Wallet Customer Lifetime Value (LTV) dataset, containing 7,000 records and 20 features covering demographic, transactional and behavioural information.
Key takeaways	1. Static behavioural information can make minority-class prediction difficult. 2. Temporal and session-based features may provide additional useful information. 3. SMOTE can improve the detection of minority-class observations in an imbalanced classification problem.

Public Dataset Used for Part B

Requirement	Information
Name	Bank Marketing dataset (bank-additional-full.csv)
Harvard reference	Moro, S., Rita, P. and Cortez, P. (2014) Bank Marketing [Dataset]. UCI Machine Learning Repository. doi: 10.24432/C5K306.
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	41,188 instances in the bank-additional-full.csv file. The older bank-full.csv version contains 45,211 instances and fewer features; the version used for this project is bank-additional-full.csv.
Number of feature columns	20 feature columns.
Number of classes	2 (yes and no).
Target range	Not applicable because this is a classification problem.
Origin	Real Portuguese bank direct-marketing campaign data collected through telephone-based campaigns. The dataset is publicly available through the UCI Machine Learning Repository.